

1. EXPLORATORY DATA ANALYSIS (EDA)

1.1 Data Collection

- Collect data from various sources (CSV, DB, APIs, etc.)
- Understand the data structure.

1.2 Data Cleaning

- Handle missing values
- Remove duplicates
- Detect & treat outliers
- Correct inconsistent data.

1.3 Data Exploration

- Understand data distribution
- Summary statistics
- Identify patterns & trends

1.4 Data Visualization

- Histograms, Bar plots
- Box plots, scatter plots
- Correlation heat maps
- Pair plots

TOOLS / LIBRARIES

- Python (Pandas, NumPy)
- Matplotlib, Seaborn, Plotly
- Script - Jupyter, Google Colab

NOTE: EDA helps in understanding the data, Model building helps in predicting & making decisions.

EXPLORATORY DATA ANALYSIS AND MODEL BUILDING



KEY BENEFITS

- Understands data deeply
- Improves data quality.
- Helps in choosing right model.
- Enhances model performance.

7. DEPLOYMENT

- Deploy model to production
- Web/Mobile/cloud Application
- Monitor model in real time.

8. FINAL PREDICTION

- Use trained model to make predictions on new data.
- Generate actionable results.

2. MODEL BUILDING

2.1 Problem Definition

- Define target variable
- Identifying the type of problem (Classification, Regression).

2.2 Feature Engineering

- Feature Selection
- Feature Transformation
- Encoding categorical variables

2.3 Train Test Split

- Split data into training & testing sets
- (e.g.: 80% Train, 20% Test)

2.4 Model Training

- Select appropriate algorithms (Linear Regression, Logistic Regression, Decision Tree)
- Train the model on training data.

2.5 Model Evaluation

- Evaluate using metrics (Accuracy, Precision, Recall, RMSE, R2 etc.)

2.6 Model Tuning

- Hyperparameters tuning
- Cross-Validation
- Improve model performance

Evaluation

- Metrics
- Accuracy
- Precision
- Recall
- F1-Score
- R2 Score

Goal

- Extract useful insights & build accurate models to support business decisions.